Spectral Variability Bayesian Unmixing for Hyperspectral Sequence in Wavelet Domain

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***Abstract*— For unmixing (UN) of the sequence of hyperspectral images (SHS), spectral variability is an important factor to be considered. However, most existing UN methods tend to model the endmember and its variability in the spatial domain rather than the transform domain. In fact, the intrinsic and invariant features of the spectral curve can be effectively represented by wavelet transform. Therefore, this article proposes to perform SHS UN in the wavelet domain by combining the Bayesian method. First, the assumption of abundance being invariability in both the spatial and wavelet domains is made, and then, the formulation of UN in the wavelet domain using the perturbed linear mixing model (PLMM) is presented. Second, based on the Bayesian framework, the likelihood and prior are both given, in which the parameter priors are divided into two parts: low- and high-frequency wavelet coefficients. Moreover, by considering the sparsity of the high-frequency wavelet coefficients of endmembers, a noninformative prior with zero mean is designed. Meanwhile, for the coefficients of endmember variability, Gaussian distributions are utilized to represent the steady fluctuation along the temporal dimension. Finally, using the maximum a posteriori (MAP) rule, a hierarchical spectral variability UN model in the wavelet domain is built and solved by the Markov chain Monte Carlo (MCMC) sampling algorithm. Numerical experiments show that the proposed method generates more accurate estimates for endmembers and their variation.**

***Index Terms*— Bayesian unmixing (UN), endmember variability, hyperspectral (HS) image, Markov chain Monte Carlo (MCMC) method.**

I. Introduction

VARIOUS successful applications with hyperspectral (HS) remote sensing images have well demonstrated the importance of observing the Earth's surface information with HS remote sensing technology [24], [58]. Meanwhile, HS unmixing (UN) has become one of the promising topics in HS remote sensing on account of the crucial influence on the downstream processing of HS image, including classification [4], and target and change detection [10], [16], [29]. HS UN aims at decomposing the HS image by a combination of spectral signatures (pure endmembers) and their coefficients (abundance) [9], [26], [40]. A variety of UN methods have been well developed, including VCA [15], [37], SISUAL [27], NMF [21], [41], sparse representation [20], and Bayesian inference [6], [13]. However, most of the existing methods assume that spectral signatures of the same materials remain unchanged in different scenes by omitting the variable imaging situation caused by illumination, atmosphere, season, and so on. Hence, techniques that can accommodate the spectral variability become an urgent issue in UN [19], [31], [32], [35], [38], [39], [42], [43]. Meanwhile, the UN of the sequence of HS images (SHS), also named the UN of multitemporal HS images, has attracted great interest because it provides temporal–spatial–spectral information of the scene [55]. However, spectral variability is one of the challenging issues to be solved in SHS UN.

By incorporating the spectral variability in the traditional linear UN model [linear mixing model (LMM)] [17], techniques for spectral bundles, also called spectral libraries, are proposed. Among them, the multiple endmember spectral mixture analysis (MESMA) is the most widely used method [11], [47], [53], in which a large library of spectra is constructed to represent the different instants of each material. Due to spectral bundles involving a high computational cost, an alternative approach was proposed to add the sparse constraints to the abundance [48]. Moreover, by combining the sparsity, the libraries of spectral and spectral variability were separated and constructed by PCA [46], which is similar to that in robust nonnegative matrix factorization (NMF) proposed in [19], but the latter is library-free UN method. In [25], a geometric description of the spectral variability by prototypal endmember spectral was used to form a library-free and bundle-driven UN model.

Another popular approach is the parametric model, which is also based on LMM. Different from spectral bundles, these methods represent the spectral variability by introducing additional parameters to the LMM model. An extended LMM (ELMM) [1], [34] assumed that the variability of each

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endmember is affected by a scaling factor learned from a physical Hapke model. However, the scaling factor for each material lacks flexibility when endmembers vary with more complex environments [28]. To address this issue, a scaling matrix was employed to represent the spectral variation of the wavelength of each individual endmember in the generalized linear mixed model (GLMM) [2], [52]. The scaling matrix of GLMM includes more parameters than that of ELMM, which results in higher computational complexity. Thus, the UN process is divided into two steps. The first is to construct a fourth low-rank tensor by pure pixels to estimate the scaling matrix, while the second step is to solve the abundance, which is a matrix-factorization problem [52]. However, the abovementioned methods depend strongly on the initial reference endmember estimated from the HS image. Hence, by considering the intraclass variability, an integrated UN chain-based ELMM was developed to extract the endmember [25]. Moreover, in order to relax the dependence of ELMM or GLMM on endmember, an additive perturbation was assumed to the LMM in the perturbed LMM (PLMM) model [7], in which the reference endmember matrix and its corresponding perturbation term are both estimated from the HS image blindly. Furthermore, one of the advantages of PLMM is that the spectral variability of each pixel may be the same or different, which results in a flexible application to multitemporal and distributed spectral UN problems [8], [36].

Different from the parametric model, the Bayesian-based statistical approach regards the abundance, endmember, and endmember variability as random variables and parameterizes their probability distributions [57]. The resulting maximum a posteriori problem can be derived by optimization approaches, e.g., expectation maximization (EM) and Markov chain Monte Carlo (MCMC). Among them, the normal compositional model (NCM) is the most popular one due to its simplicity [13], which models an endmember as a multivariate Gaussian distribution and estimates the parameter of each endmember under a Bayesian framework by EM iteration. Taking the nonconvexity of EM into consideration, NCM was improved by utilizing MCMC to sample the posterior probability and setting a conjugate inverse gamma (IG) distribution to the endmember variance [13]. A beta compositional model (BCM) [14] was developed to avoid negative estimates in Gaussian distribution. In addition, a multimodal distribution, the Gaussian mixture model (GMM), was proposed to model the complexity of the endmember distribution [50]. The parameters in GMM may be learned in a supervised or unsupervised way with the latter produced in a clustered endmember set. Recently, deep generative modes (DGMs), such as generative adversarial networks (GANs) and variational autoencoders (VAEs), have shown remarkable performance for modeling complex prior distributions form limited samples [56]. In [36], they combined the Bayesian inference with neural networks (NNs) to UN [probabilistic generative model for spectral UN (PGMSU)], in which a probabilistic generative model is used to address the endmember variability, and the NNs are applied to model the nonlinearity. Furthermore, a VAE-based deep generative model for UN was also proposed in [54] with a more flexible superpixelwise manner.

Compared with the HS UN, the UN of SHS is more complex with a variation in temporal dimension, which involves more parameters and issues to be considered [18], [30], [33]. In [55], the high temporal correlation between the abundances for UN and change-detection in SHS data was studied. The abundance was modeled as a combination of three parts, a small additive term, a sparsity term, and a high-magnitude term; then, the abrupt changes in the time domain can be detected. Some of the UN methods for SHS assume that the spectra are varied with time. Hence, the endmember dictionary is pruned by the fact that the HSI data lie in a low-dimensional subspace, and then, a collaborative sparse UN was followed [45]. Most of the studies develop a parametric model of single HSI UN to SHS UN by considering the spatial and temporal or endmember and abundance variations based on the LMM, the spectral variability is modeled as a scale-additive term, and the abundance has a potential change in spatial distribution [44]. Furthermore, the model was interpreted from two perspectives: one is tensor decomposition and the other is the standard state-space model. Following the state-space view in [44], the endmember and abundance were set as state variables in the dynamical equation, and an NN was employed to learn the dynamical systems [51]. In [8], the parameters of the PLMM were estimated by a two-stage stochastic program online, while, in [5], the UN and change-detection were performed at the same time by combining the PLMM and Bayesian framework, in which the stochastic variables, including abundance, endmember, endmember variability, and noise, are set with different probability distributions, respectively. Specifically, the smoothness of the abundance and small perturbation of the spectral variability in the temporal domain were modeled as Gaussian distributions. Furthermore, the spatial correlation and the spectral smoothness of endmember variability in SHS were taken into consideration to achieve a more accurate estimation for spectral variability [22], [23]. Although these Bayesian-based UN methods achieve satisfying UN results, the prior distributions of the endmember and its variability have been modeled by addressing the spatial or temporal relationship, which omits the inherent features of the spectral curve itself.

It is well-known that wavelet transform has a great advantage to represent curves. Specifically, the low and high frequencies of the wavelet coefficients represent the trend and details of the curve, respectively. When it comes to UN, the low-frequency coefficient for the endmember curves stands for the invariable part, and the high-frequency coefficient may give detail information (extreme or inflection points and so on) of the spectral curves, which is related to the inherent absorption feature of the material. When it confines to SHS UN, the intrinsic feature of endmember spectra should be the same no matter how the imaging environment changes. Hence, it is more reliable and accurate to study the spectral curve in the wavelet domain [12]. Inspired by this observation, the wavelet coefficient priors of the endmember and its variability are fully exploited under the Bayesian framework for SHS

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UN. The main contributions of this article are presented as follows.

First, on account that the proportion of the material in the scene is invariable, the abundance in the spatial or transform domain is assumed as a constant. Then, the formulation of the UN model based on PLMM in the wavelet domain is given. Moreover, the parameter priors in the wavelet domain include endmembers, endmember variability, and noise, as well as hyperparameters, which are constructed according to their properties with various probability distributions. Finally, a joint posterior probability is derived, and the Gibbs sampler is employed to implement the Bayesian inference. This article is organized as follows. The PLMM UN model is introduced in Section II. In Section III, the wavelet-based UN model is proposed, and the prior of wavelet coefficients of endmembers and their variability are developed. Section IV investigates a Gibbs sampler to solve the resulting UN model. Section V shows the performance of the proposed approach on synthetic and real data compared with state-of-the-art UN approaches. Finally, Section VI concludes this work.

II. PLMM

With the consideration of the spectral variability, PLMM is represented as follows:

$$\mathbf{Y} = (\mathbf{M} + \mathbf{dM})\mathbf{A} + \mathbf{B} \tag{1}$$

where $\mathbf{Y} \in \mathbb{R}^{L\times N}$ is an HS image with L bands and N pixels, $\mathbf{M} = [\mathbf{m_1}, \mathbf{m_2}, \ldots, \mathbf{m_R}] \in \mathbb{R}^{L\times R}$ is the endmember matrix with R pure materials (or endmembers) presented in the observed scene, $\mathbf{A} \in \mathbb{R}^{L\times R}$ is the abundance, $\mathbf{dM}$ is the variation part of spectra with the same size as M and a small perturbation d, and $\mathbf{B} \in \mathbb{R}^{L\times N}$ represents the Gaussian noise.

The observed SHS $\mathbf{Y}_t$ is acquired at T different instants, which the same endmembers, i.e., $\mathbf{M} = \mathbf{M}_t$. Then, $\mathbf{Y}_t$ can be modeled as [7]

$$\mathbf{Y}_t = (\mathbf{M}_t + \mathbf{dM})\mathbf{A}_t + \mathbf{B}_t \tag{2}$$

where $t = 1, 2, \ldots, T$. Each $\mathbf{Y}_t = [\mathbf{y}_{1,t}, \ldots, \mathbf{y}_{N,t}]$ is an $L \times N$ matrix, $\mathbf{A}_t$ is composed of the abundance vector $a_{n,t}$ at time t with $\mathbf{A}_t = [\mathbf{a}_{1,t}, \ldots, \mathbf{a}_{N,t}]$, and $\mathbf{dM}_t = [\mathbf{dm}_{1,t}, \ldots, \mathbf{dm}_{R,t}]$ is an $L \times N$ matrix accounting for the noise at time t.

III. Proposed Method

A. Unmixing in Wavelet Domain

Performing 1-D wavelet transform along the spectral dimension of HSI, the PLMM UN model in the wavelet domain can be represented as

$$\mathrm{wt}(\mathbf{Y}_t) = \mathrm{wt}[(\mathbf{M} + \mathbf{dM}_t) * \mathbf{A}_t + \mathbf{B_t}] \tag{3}$$

where $\mathrm{wt}(\cdot)$ represents the wavelet transform operator.

Furthermore, due to the linearity of the wavelet operator, (3) can be written as

$$\mathrm{wt}(\mathbf{Y}_t) = [\mathrm{wt}(\mathbf{M}) + \mathrm{wt}(\mathbf{dM}_t)] * \mathbf{A}_t + \mathrm{wt}(\mathbf{B}_t). \tag{4}$$

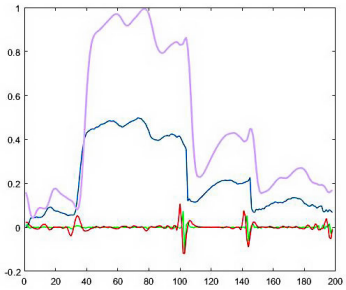


Fig. 1. Wavelet transform coefficients of spectral curves.

It is assumed that the abundance coefficients in the wavelet domain are the same as those in the spatial domain. The physical interpretation is that the distribution of ground materials is unchanged, and their proportions are fixed, so the abundance coefficients should be constant no matter in the spatial or wavelet domain. Therefore, the UN model in the wavelet domain just needs to pay attention to endmembers, endmember variability, and noise.

In the Bayesian framework, the wavelet domain UN based on PLMM is represented as the problem of maximizing the joint posterior distribution

$$\begin{aligned}&\max_{\Theta} p(\Theta \mid \mathrm{wt}(\mathbf{Y}_t))\\ &= p((\mathbf{wt}(\mathbf{Y}_t \mid \Theta))p(\mathbf{wt}(\mathbf{M}))\\ &\quad * \frac{p\left(\mathrm{wt}\left(\mathbf{dM}_t \mid \mathbf{wt}(\mathbf{M}), \Psi_w^2\right)\right)p(\mathbf{A}_t)p\left(\sigma_t^2\right)p\left(\mathbf{\Psi}_w^2\right)}{p(\mathbf{wt}(\mathbf{Y}_t))}\end{aligned} \tag{5}$$

where $\Theta = \{\mathrm{wt}(\mathbf{M}), \mathrm{wt}(\mathbf{dM}_t), \mathbf{A}_t, \sigma_t^2, \mathbf{\Psi}_w^2\}$ represents the parameters and hyperparameters, σ_t^2 is the wavelet-domain noise variance, $\mathbf{\Psi}_w^2$ is the wavelet-domain variance of the endmember variability, and $p(\mathrm{wt}(\mathbf{Y}_t)|\Theta)$ is the likelihood function.

B. Bayesian Unmixing Model in Wavelet Domain

The likelihood function $p(\mathrm{wt}(\mathbf{Y}_t)|\Theta)$ in the wavelet domain can be expressed as

$$\begin{aligned}p(\mathrm{wt}(\mathbf{Y}_t) \mid \Theta) \propto &\prod_{t=1}^{T}\left(\sigma_t^2\right)^{-\frac{(\lambda+1)NL}{2}}\\ &* \exp\left(\frac{-1}{2\sigma_t^2}\| \mathrm{wt}(\mathbf{Y}_t) - (\mathrm{wt}(\mathbf{M}) + \mathrm{wt}(\mathbf{dM}_t))\mathbf{A}_t\|_F^2\right)\right)\end{aligned} \tag{6}$$

where $\|\cdot\|_F$ represents the Frobenius norm, λ represents the scale of the wavelet transform, N is the number of the pixels, and L is the number of bands.

1) Endmember Prior in Wavelet Domain: Fig. 1 gives the wavelet coefficients of an endmember curve after two-scale wavelet decompositions. The blue one stands for the spectral curve, while the purple one represents its low-frequency wavelet coefficient. It can be seen that the low-frequency component can approximate the curve. However, before UN, endmembers are unknown and so does the basic trend. In this case, in order to have as little influence on the posterior distribution as possible, a noninformative prior is assumed. Namely, let the data speak for themselves [3]. Therefore,

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noninformative prior is used as the prior of the low-frequency coefficient of the endmember $\mathrm{wt}(m)^{\mathrm{low}}$

$$\mathrm{wt}(m)_r^{\mathrm{low}} \sim N\left(0, \xi^{\mathrm{low}}\right) \tag{7}$$

where $\mathrm{wt}(m)_r^{\mathrm{low}}$ represents the low-frequency data of the rth endmember and ξ^{low} is a sufficiently large number to ensure noninformative prior.

In Fig. 1, the red and green curves represent its high-frequency wavelet coefficients in two scales. It can be seen that the high-frequency components represent the details, including peaks and valleys with sparsity. Furthermore, the modulus maxima of the high frequency can well reflect the information of sharp change (absorption band) of endmembers in some bands, which are their inherent characteristic [12]. Moreover, the spectral absorption characteristics of endmember curves have an important impact on UN results. By extracting the modulus maxima, the position of the peak and valley value of the curve can be well located. Thus, the first k modulus maxima of the wavelet coefficients, denoted as $\mathrm{wt}(m_r)_k^{\mathrm{high}}$, are extracted first, and then, the prior of the modulus maxima is estimated to approximate to the prior of the high-frequency coefficients. However, since the location of the modulus maximum is unknown, noninformative prior is also used to model the modulus maximum of the high-frequency coefficients

$$\mathrm{wt}(m_k)_r^{\mathrm{high}} \sim N\left(0, \xi^{\mathrm{high}}\right) \tag{8}$$

where $\mathrm{wt}(m_k)_r^{\mathrm{high}}$ represents the kth modulus maximum of the high-frequency data of the rth endmember and ξ^{high} is a sufficiently large number to ensure noninformative prior.

2) Endmember Variability Prior in Wavelet Domain: In PLMM, it is assumed that the endmember variability is a small perturbation, which obeys the Gaussian distribution with small variance and zero mean [5]. In the wavelet domain, the corresponding wavelet coefficients are set in the same distribution as that of the spatial domain.

Therefore, the prior of the low-frequency wavelet coefficients of the rth endmember variability at $t = 1$ is

$$\mathrm{wt}(\mathbf{dm})_{r,1}^{\mathrm{low}} \sim N\left(0, v^{\mathrm{low}}\right). \tag{9}$$

For the high-frequency wavelet coefficients of endmembers variability, the first k modulus maximum of the high- frequency of the rth endmember variability coefficient at $t = 1$ can be written as

$$\mathrm{wt}(\mathbf{dm}_k)_{r,1}^{\mathrm{high}} \sim N\left(0, v^{\mathrm{high}}\right). \tag{10}$$

It should be mentioned that both of the low- and high-frequency variances v^{low} and v^{high} have every small values. Furthermore, v^{high} is smaller than v^{low}, which means that the high frequency is a finer detail than the low frequency does.

In terms of time, considering the stable variation of endmember variability, the current time is generated based on the previous time adding a small disturbance, so

$$\mathrm{wt}(\mathrm{dm})_{l,r,t}^{\mathrm{low}} | \mathrm{wt}(\mathrm{dm})_{l,r,t-1}^{\mathrm{low}} \sim N\left(\mathrm{wt}(\mathrm{dm})_{l,r,t-1}^{\mathrm{low}}, \left(\psi_{l,r}^2\right)_w^{\mathrm{low}}\right)$$

$$\mathrm{wt}(\mathrm{dm}_k)_{l,r,t}^{\mathrm{high}} | \mathrm{wt}(\mathrm{dm}_k)_{l,r,t-1}^{\mathrm{high}} \sim N\left(\mathrm{wt}(\mathrm{dm}_k)_{l,r,t-1}^{\mathrm{high}}, \left(\psi_{l,r}^2\right)_w^{\mathrm{high}}\right) \tag{11}$$

where $\mathrm{wt}(\mathrm{dm})_{l,r,t}^{\mathrm{low}}$ represents the low-frequency data of endmember variability in the wavelet domain, $\mathrm{wt}(\mathrm{dm}_k)_{l,r,t}^{\mathrm{high}}$ represents the first k modulus maximum of the high frequency of the rth endmember variability coefficient in the l-band at t instance, and $(\psi_{l,r}^2)_w^{\mathrm{low}}$ and $(\psi_{l,r}^2)_w^{\mathrm{high}}$ represent the variance of high- and low-frequency data, respectively.

3) Abundance Prior in Wavelet Domain: According to (4), the abundance in the wavelet domain is the same as that of the spatial domain. Thus, the sum-to-one and nonnegativity of abundance constraints are held on. Furthermore, the consistency prior in [23] is also introduced as follows.

For $t = 1$, the uniform distribution is used as a prior distribution of the first moment of the abundance

$$\mathbf{a}_{n,1} \sim U_{\mathbf{S_R}} \tag{12}$$

where $U_{\mathbf{S_R}}$ denotes the uniform distribution on the set $\mathbf{S_R}$ with

$$\mathbf{S_R} = \{\mathbf{x} \in R^R | \forall i, x_i \geq 0, \mathbf{x}^T \mathbf{l}_R = 1\}. \tag{13}$$

For $t > 1$

$$\begin{aligned} p\left(a_{r,n,t} | A_{\setminus\{a_{r,n,t}\}}\right) &\propto \exp\left(\frac{-1}{2\varepsilon_n^2} \|a_{r,n,t} - a_{r,n,t-1}\|_2^2\right. \\ &\quad \left. * \exp\left\{\frac{-1}{2\sigma^2\left(a_{r,n,t}\right)} \|a_{r,n,t} - \bar{a}_{r,n,t-1}\|_2^2\right\} \mathbf{l}_{S_R}\left(a_{n,t}\right)\right. \end{aligned} \tag{14}$$

where ε_n^2 is the variance, and usually, it is a small fixed value (e.g., $\varepsilon_n^2 = 10^{-3}$); $\mathbf{l}_{\mathbf{S_R}}$ is the indicator function of the set $\mathbf{S_R}$; and the standard variance $\sigma(a_{r,n,t})$ is

$$\sigma\left(a_{r,n,t}\right) = \alpha \sqrt{\frac{\left(\nabla_{n,t}^h\left(a_{r,n,t}\right)\right)^2 + \left(\nabla_{r,n,t}^v\left(a_{r,n,t}\right)\right)^2}{2}} \tag{15}$$

with the horizontal and vertical first-order differences operators being $\nabla_{n,t}^h(a_{r,n,t})$ and $\nabla_{r,n,t}^v(a_{r,n,t})$, respectively, and α being used to adjust the probability.

4) Hyperparameter Prior in Wavelet Domain: Similarly, as in the spatial domain, the prior of noise variance σ_t^2 and variability variance Ψ^2 are set as inverse-Gamma prior, which is a conjugate prior of the Gaussian distribution

$$\sigma_t^2 \sim \mathrm{IG}(a_\sigma, b_\sigma) \tag{16}$$

$$\left(\Psi_{l,r}^2\right)_w^{\mathrm{low}} \sim \mathrm{IG}\left(a_\Psi^{\mathrm{low}}, b_\Psi^{\mathrm{low}}\right) \tag{17}$$

$$\left(\Psi_{l,r}^2\right)_w^{\mathrm{high}} \sim \mathrm{IG}\left(a_\Psi^{\mathrm{high}}, b_\Psi^{\mathrm{high}}\right) \tag{18}$$

where parameters $a_\sigma, b_\sigma, a_\Psi^{\mathrm{low}}, b_\Psi^{\mathrm{low}}, a_\Psi^{\mathrm{high}}$, and b_Ψ^{high} are the corresponding hyperparameters of the IG distribution, and they are sufficiently small in order to ensure a weakly informative prior.

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Algorithm 1 Gibbs Sampler

Data:burn-in iterations N_{bi}, maximum iteration number N_{MC}, $M^{(0)}$, $A^{(0)}$, $dM^{(0)}$, $(\sigma^2)^{(0)}$, $(\psi^2)^{(0)}$
Begin
For $q = 1 : N_{MC}$
Sampling the endmembers $\mathbf{M}^{(q)}$ according to Eq.(20);
Sampling the endmember variability terms $\mathbf{dM}^{(q)}$ according to Eq.(23);
Sampling the abundances $\mathbf{A}^{(q)}$ according to Eq.(26);
Sampling the noise variances $\sigma^{2(q)}$ according to Eq.(29);
Sampling the variability variances $\Psi^{2(q)}$ according to Eq.(30)
End
Result:$\{\mathbf{M}^{(q)}, \mathbf{dM}^{(q)}, \mathbf{A}^{(q)}, \sigma^{2(q)}, \psi^{2(q)}\}_{q=1}^{N_{MC}}$

IV. Gibbs Sampler Strategy for SHS Unmixing

With the prior distribution of endmembers and their variability, the posterior distribution of the parameters in the wavelet domain can be obtained by the Bayesian formula [5], as shown in (19), bottom of the page.

An MCMC method is applied to sample the posterior (19) to estimate the parameters $\mathbf{M}, \mathbf{dM}_t$ and $\mathbf{A}_t$ with the minimum mean square error (MMSE) estimator after $\mathbf{N}_{bi}$ burn-in iterations. The main steps of the proposed Gibbs sampler are summarized in Algorithm 1.

A. Sampling the Endmember M

$$\mathrm{wt}(m)_{l,r}|\mathrm{wt}(\mathbf{Y}), \mathbf{\Theta}_{\{\mathrm{wt}(m)_{l,r}\}} \sim \mathrm{N}\left(\mu_{l,r}^{(\mathbf{M})}, \kappa_{l,r}^2\right) \tag{20}$$

$$\mu_{l,r}^{(\mathbf{M})} = \kappa_{l,r}^2 \sum_{t=1} \frac{-1}{2\sigma_t^2}[\mathrm{wt}(\widetilde{\mathbf{y}})_{1,t} - \mathrm{wt}(\widetilde{\mathbf{m}})_{1,\backslash r}\mathbf{A}_{\backslash r,t} - \mathrm{wt}(d\widetilde{\mathbf{m}})_{1,t}\mathbf{A}_t]\tilde{a}_{r,t} \tag{21}$$

$$\kappa_{l,r}^2 = \left[\sum_{n,t} \frac{a_{r,n,t}^2}{\sigma_t^2} + \frac{1}{\xi}\right]^{-1}. \tag{22}$$

B. Sampling the Endmember Variability dM

Similarly, the combination of the likelihood function (5) and the prior (9)–(11) leads to

$$\mathrm{wt}(\mathrm{dm})_{l,r,t} \sim \mathrm{N}\left(\mu_{l,r,t}^{(\mathbf{dM})}, \eta_{l,r,t}^2\right) \tag{23}$$

$$\frac{1}{\eta_{l,r,t}^2} = \frac{1}{\sigma_t^2}\sum_n a_{r,n,t}^2 + \frac{1}{v}\delta(t=1) + \frac{1}{\psi_{l,r}^2}(1 + \delta(1 < t < T) \tag{24}$$

$$\mu_{l,r,t}^{(\mathbf{dM})} = \left[\frac{1}{\sigma_t^2}\left((\widetilde{\mathbf{y}})_{l,t} - \mathrm{wt}(\boldsymbol{d}\widetilde{\boldsymbol{m}})_{l,\backslash,r,t}\boldsymbol{A}_{\backslash r,t}\right) - \mathrm{wt}(\widetilde{\boldsymbol{m}})_l \boldsymbol{a}_{n,t}\right]\tilde{a}_{n,t}^T$$
$$+ \frac{1}{\left(\psi_{l,r}^2\right)_w}\left(\delta(t < T)\mathrm{wt}(\mathbf{dm})_{l,r,t+1} + \delta(t > 1)\mathrm{wt}(\mathbf{dm})_{l,r,t-1}\right)\eta_{l,r,t}^2\Bigg]. \tag{25}$$

C. Sampling the Abundance A

The likelihood function (5) combined with the prior (12)–(15) leads to the following conditional distribution for the abundances (27) and (28), as shown at the bottom of the next page

$$\boldsymbol{a}_{n,t}|\mathbf{y}_{n,t}, \mathbf{\Theta}_{\{\boldsymbol{a}_{n,t}\}} \sim N_{\mathbf{S_R}}\left(\mu_{n,t}^{(A)}, \mathbf{\Lambda}_{n,t}\right). \tag{26}$$

where $\delta(\cdot)$ represents an indicator function and $N_{\mathbf{S_R}}(\mu, \mathbf{\Lambda})$ denotes a Gaussian distribution truncated to the set $\mathbf{S_R}$.

D. Sampling the Variances of Noise σ^2 and Variability Ψ^2

Using the likelihood function (5) and priors (15) and (16), the sampled noise and variability variances are represented as

$$\sigma_t^2|\mathrm{wt}(\mathbf{Y}_t), \Theta_{\{\sigma_t^2\}} \sim \mathrm{IG}\left(\mathrm{a}_\sigma + \frac{\mathrm{LN}}{2},\ \mathrm{b}_\sigma + \frac{1}{2}\|\mathrm{wt}(\mathbf{Y}_t) - (\mathrm{wt}(\mathbf{M}) + \mathrm{wt}(\mathbf{dM}_t))\mathbf{A}_t\|_F^2\right) \tag{29}$$

$$\left(\Psi_{l,\mathrm{r}}^2\right)_{\mathrm{W}} \mid \Theta_{\{\psi_{l,\mathrm{r}}^2\}} \sim \mathrm{IG}\left(\mathrm{a}_\psi + \frac{\mathrm{T}-1}{2},\ \mathrm{b}_\psi + \frac{1}{2}\sum_{t=2}^{\mathrm{T}}\left(\mathrm{wt}(\mathrm{dm})_{l,\mathrm{r},t} - \mathrm{wt}(\mathrm{dm})_{l,\mathrm{r},t-1}\right)^2\right). \tag{30}$$

The Gibbs sampler in the MCMC method is used to sample and calculate the posterior distribution of parameters. The algorithm is summarized in Algorithm 1.

It should be noted that Algorithm 1 is sampled in the wavelet domain, while, in our previous work [23], the sampling is performed in the spatial domain. What is more, by integrating the temporal–spatial–spectral dimension information, this article stresses the endmember and its variability wavelet coefficient distributions; the model in [23] mainly focuses on the abundance and endmember variability distributions.

V. Experiments

In this section, the experiments on simulated and real datasets are conducted. The proposed WT_HB method is compared with VCA/FCLS [15], SISAL/FCLS [27], PLMM [7], ELMM [1], online UN (OU) [8], hierarchical Bayesian (HB) [5], mixed prior of BayesianMI (MIX_HB) [23], and PGMSU [36]. Among them, VCA/FCLS and SISAL/FCLS are classical UN without considering the spectral variability. Others taking the spectral variability into account, i.e., PLMM

$$p\left(\mathrm{wt}(\mathbf{M}) \mid \mathrm{wt}(\mathbf{Y}_t), \Theta_{\backslash \mathrm{wt}(M)}\right) p\left(\mathbf{A}_t \mid \mathrm{wt}(\mathbf{Y}), \Theta_{\backslash A_t}\right) * p\left(\mathrm{wt}(\mathbf{dM}_t) \mid \mathrm{wt}(\mathbf{Y}), \Theta_{\backslash \mathrm{wt}(dM_t)}\right) p\left(\sigma_t^2 \mid \mathrm{wt}(\mathbf{Y}), \Theta_{\backslash \sigma_t^2}\right) p\left(\Psi_{\mathbf{w}}^2 \mid \Theta_{\backslash \Psi^2}\right)$$
$$\propto p(\mathrm{wt}(\mathbf{Y_t}) \mid \Theta) p\left(\mathrm{wt}(\mathbf{dM}_t) \mid \mathrm{wt}(\mathbf{M}), \Psi_{\mathbf{w}}^2\right) * p(\mathrm{wt}(\mathbf{M})) p(\mathbf{A}_t) p\left(\sigma_t^2\right) p\left(\Psi_{\mathbf{w}}^2\right)\Big) \tag{19}$$

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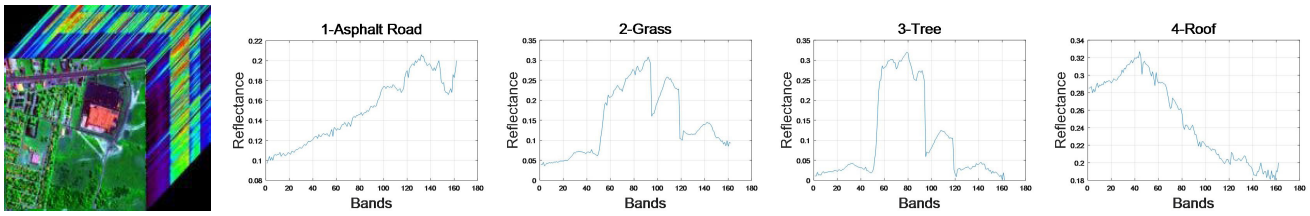
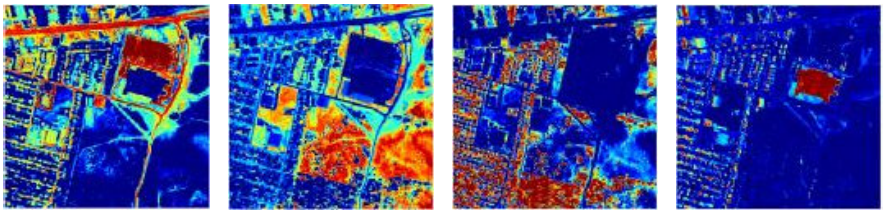


Fig. 2. Urban dataset and the reference four endmember curves.



(a) (b) (c) (d)

Fig. 3. Corresponding abundance maps of the Urban dataset with four endmembers. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof.

TABLE I
ASSESSMENT INDEXES

Indexes	Definitions
aSAM	$aSAM(\mathbf{M}) = \frac{1}{R}\sum_{r=1}^{R} \arccos(\frac{m_r^T \hat{m}_r}{\|m_r\|_2 \|\hat{m}_r\|_2})$
GMSE(A)	$GMSE(\mathbf{A}) = \frac{1}{\text{TRN}}\sum_{t=1}^{T} \|\mathbf{A}_t - \hat{\mathbf{A}}_t\|_F^2$
GMSE(dM)	$GMSE(\mathbf{dM}) = \frac{1}{\text{TLR}}\sum_{t=1}^{T} \|\mathbf{dM}_t - \mathbf{d}\,\hat{\mathbf{M}}_t\|_F^2$
RE	$RE = \frac{1}{\text{TLN}}\sum_{t=1}^{T} \|\mathbf{Y}_t - \mathbf{d}\hat{\mathbf{Y}}_t\|_F^2$

and ELMM, are the UN models considering the spectral variability based on LMM, and the OU method is an improved PLMM, while the HB and MIX_HB methods analyze the SHS UN problem from the point of Bayesian view, and the PGMSU mixed the Bayesian and NNs.

Four evaluation indexes are shown in Table I. The endmember evaluation is the average of Spectral Angle Mapper (aSAM), which measures the similarity of the endmember $\mathbf{m}_r$ and the estimated endmember $\hat{\mathbf{m}}_r$. The assessment of the abundance and endmember variability are given by their mean square error (mse), namely, the global mse (GMSE) of abundance [GMSE(A)] and variability [GMSE(dM)]. In addition, the reconstruction errors (REs) of the true and estimated SHS are also used to give a general assessment. It is clear that the smaller the indexes are, the better the performance is.

A. Synthetic Data

1) Data Generation: To verify the effectiveness of the proposed method, two HS image datasets are employed. One is the Urban dataset with a size of 307 × 307 × 162, which provides the ground truth of six different endmembers (*Asphalt Road, Grass, Tree, Roof, Metal*, and *Dirt*) and their corresponding abundances. Figs. 2 and 3 show the true HSI and four endmembers, as well as their abundances. Another dataset is the Jasper dataset with size of 512 × 614 × 224, which provides the ground truth of four different endmembers (*Road, Soil, Water*, and *Tree*), and their corresponding abundances are shown in Figs. 4 and 5.

Synthetic SHS is produced in the following way. First, the known abundance map is regarded as a reference at $t = 1$, and other abundances are generated by multiplying the reference maps with trigonometric functions (e.g., $\cos((\pi/100) + t(36\pi/100))$ and $\sin((\pi/100) + t(36\pi/100))$, with $t = 1, \ldots, T\,(T = 6)$ to ensure that the evolution of time is smooth enough. In addition, the abundance at t=6 must meet the sum-to-one constraint. Then, the endmember variability is produced by multiplying the reference endmembers with randomly piecewise-affine functions, as mentioned in [5]. Finally, the synthetic SHS is generated with a 25-dB signal-to-noise ratio with the same noise variance according to the following noise variance analysis.

2) Parameters' Setting: The parameters in other methods refer to their original setting, while, for our method, they are adjusted manually for optimum in terms of aSAM of endmember and GMSE(A) of abundance, simultaneously. Moreover, to alleviate the effects of random sampling, the final result is the average of multiple iterations, which also benefits the setting of parameters and hyperparameters.

$$\mathbf{\Lambda}_{n,t}^{-1} = \frac{1}{\sigma_t^2}(\text{wt}(\mathbf{M}) + \text{wt}(\mathbf{dM}_t))^T(\text{wt}(\mathbf{M}) + \text{wt}(\mathbf{dM}_t)) + \frac{1}{\varepsilon_{n,t}^2}(\delta(t > 1) + \delta(t < T))\mathbf{I}_R + \text{diag}\left(\frac{1}{\sigma^2(a_{1,n,t})}, \ldots, \frac{1}{\sigma^2(a_{R,n,t})}\right) \tag{27}$$

$$\mu_{n,t}^{(A)} = \mathbf{\Lambda}_{n,t}\left[\frac{1}{\sigma_t^2}(\text{wt}(\mathbf{M}) + \text{wt}(\mathbf{dM}_t))^T \boldsymbol{y}_{n,t} + \frac{1}{\varepsilon_{n,t}^2}(\delta(t > 1)\boldsymbol{a}_{n,t-1} + \delta(t < T)\boldsymbol{a}_{n,t+1}) + \left[\frac{\bar{a}_{1,n,t}}{\sigma^2(a_{1,n,t})}, \ldots, \frac{\bar{a}_{R,n,t}}{\sigma^2(a_{R,n,t})}\right]^T \right. \tag{28}$$

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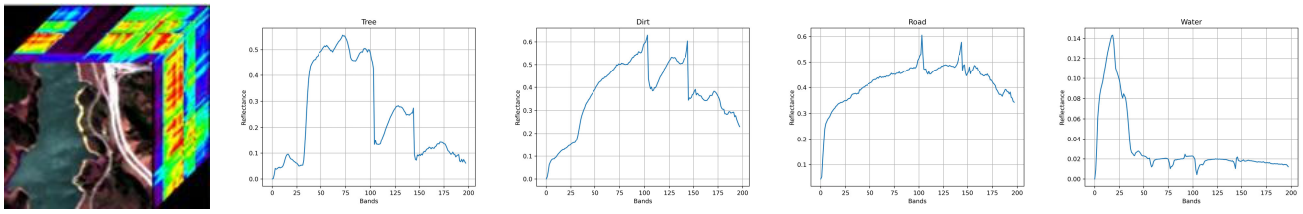
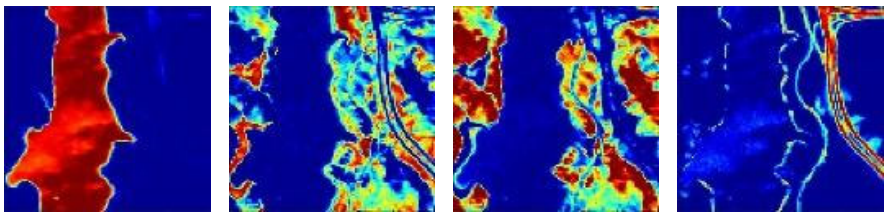


Fig. 4. Jasper dataset and the reference four endmember curves.



(a) (b) (c) (d)

Fig. 5. Corresponding abundance maps of the Jasper dataset with four endmembers. (a) Water. (b) Tree. (c) Tree. (d) Road.

TABLE II
FIXED PARAMETERS AND INITIAL VARIABLES

Parameters	Synthetic data	Real data
ε_n^2	10^{-3}	10^{-2}
ξ^{low}, ξ^{high}	1, 0.3	1, 0.3
ν^{low}, ν^{high}	10^{-6}, 10^{-8}	10^{-7}, 10^{-8}
a_ψ, a_σ	10^{-4}	10^{-4}
b_ψ, b_σ	10^{-5}	10^{-5}
N_{bi}	250	250
N_{MC}	300	300
σ_t^2	10^{-4}	10^{-4}
$(\psi_{l,r}^2)_w^{low}, (\psi_{l,r}^2)_w^{low}$	10^{-6}	10^{-6}
k	10	10
α	1.2	1

TABLE III
QUALITY INDEXES ON THE VARIANCE OF NOISES

Cases	Indexes	Same-Var	Diff-Var
Case 1	aSAM(M)	**4.51**	4.73
	GMSE(A)	**0.0164**	0.0170
	GMSE(dM)	**0.0005**	0.0006
	RE	**5.79e-5**	5.93e-5
Case 2	aSAM(M)	4.3070	**4.2218**
	GMSE(A)	0.0095	**0.0067**
	GMSE(dM)	**0.0003**	0.0004
	RE	5.99e-5	**5.98e-5**

Specifically, in the Bayesian-based UN methods, HB, MIX_HB, and WT_HB, the common parameters shown in Table II are set as that of HB. Although Bayesian-based methods (HB, MIX_HB, and WT_HB) have more than ten undetermined parameters, they will not produce major instability in UN, which is due to better generalization of the Bayesian-based methods. More details about the parameters and hyperparameters are given in [5] and [23]. Here, noise variance is taken into consideration as follows.

Take Urban SHS with four endmembers for example. First, the SHS dataset is added with Gaussian noise in two cases: case 1 is with the same variances and case 2 is with different variances. Then, the proposed WT_HB UN method is performed with the same or different variances. Let Same-Var stands for the noise variance with the same distribution, as shown in (15) and (27). Diff-Var represents the different independent variances σ_{ii}^2 that are given to each ith band. Moreover, when the variance is different, the IG distribution is employed as prior, i.e., $\sigma_{ii}^2 \sim \text{IG}(1, (\gamma_i/2))$; the hyperparameter γ_i is set with noninformation prior, and its posterior distribution is $\gamma_i \sim \text{IG}(\text{TN}+1, (TN/2))$, which is different from (15) and (27).

Table III gives the assessment indexes for the two cases. It can be seen that, when the noise prior is consistent with the real case, it performs better. On the other hand, when the noise prior is mismatched with the real one, both of the two methods get lower values. However, the difference between these two noise priors is slightly weak, which means the stability of the method to noise levels. Hence, in the following experiments, case 1 is adopted for simplicity.

3) Experimental Results: Since the ground truth of the Urban dataset provides multiple endmember numbers, UN experiments with four, five, and six endmembers are conducted. The quantitative indexes for Urban and Jasper datasets are shown in Tables II and III, respectively. The endmember curves and abundance maps for both datasets are shown in Figs. 6–9.

4) Endmember and Its Variability Estimation: From Figs. 6 and 7, it can be seen that, as expected, VCA and SISAL perform poorly in the scenario, which involves spectral variability with scattered curves. Moreover, there are some negative values that occurred in SISAL, which may be owe to the absence of the nonnegative constraint in SISAL. The performance of PGMSU is better than those of VCA and SISAL in most cases, but it performs worse for EM.4 in Fig. 6 and EM.3 and EM.4 in Fig. 7, which exhibits big deviation from other curves. The varied curves of PLMM and ELMM are too compressed to show the spectral variation, that can be owe to minimize the variance of the endmember variability. In addition, most of the curves are only shifted with the original one, which cannot

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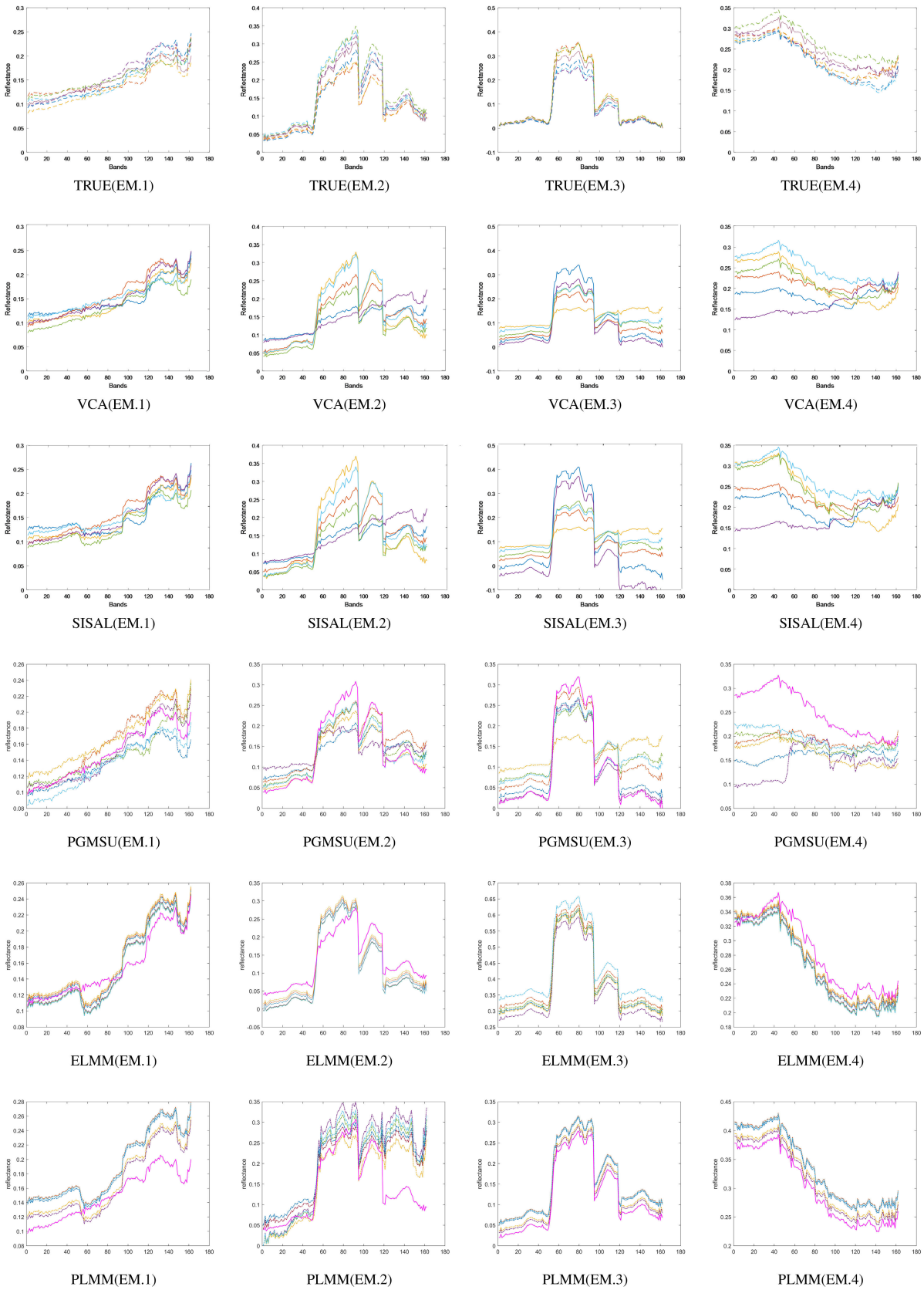


Fig. 6. Endmembers (m_r, purple lines) and their endmember variability ($m_r + dm_{r,t}$, dotted lines) recovered from the Urban dataset by different methods.

fully exhibit the multitemporal spectral variation. Compared with PLMM, the OU method has an improvement on the variation of the endmember with more dash lines in the figures, especially for the Jasper dataset. However, there are

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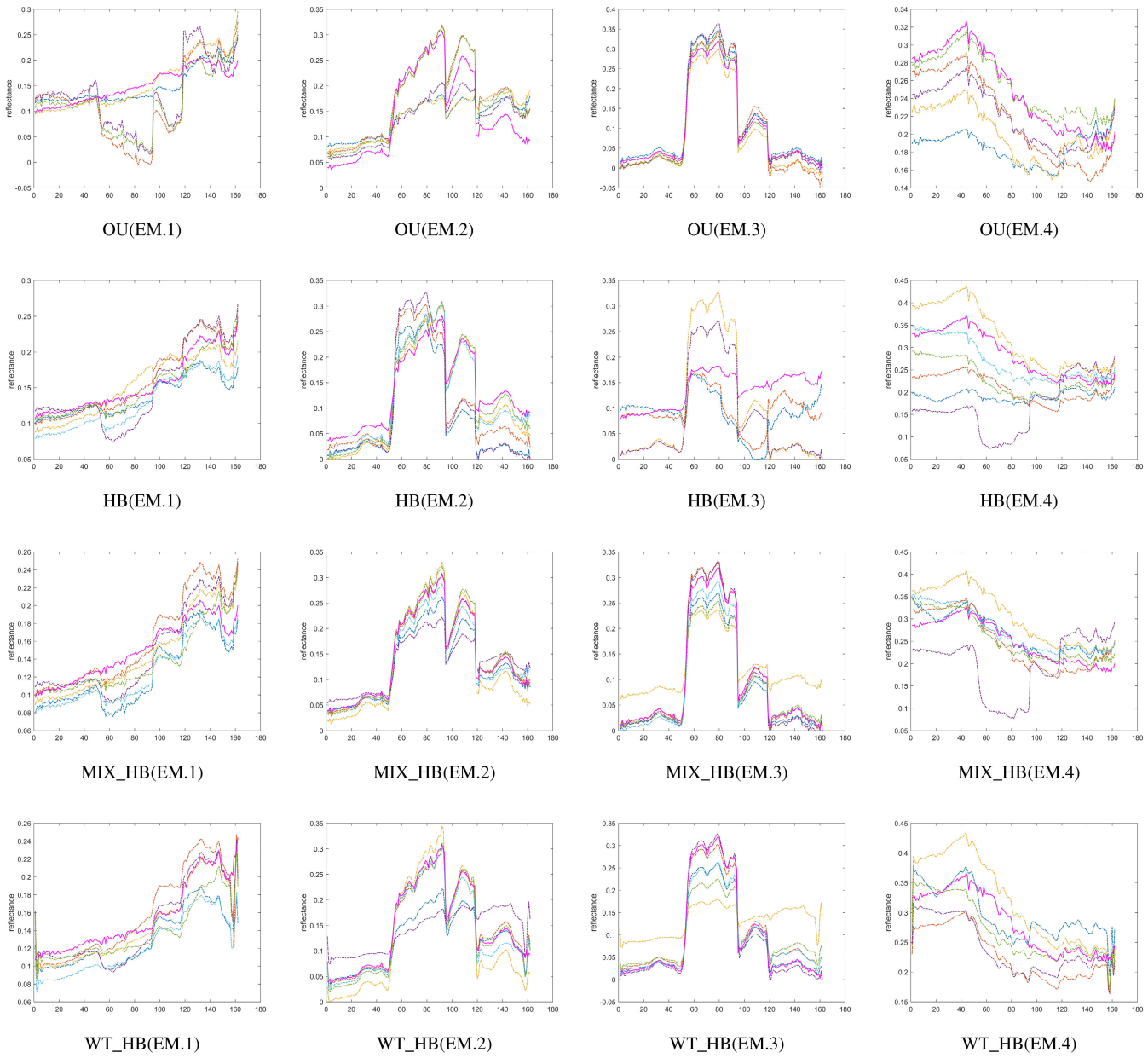


Fig. 6. *(Continued.)* Endmembers (m_r, purple lines) and their endmember variability ($m_r + dm_{r,t}$, dotted lines) recovered from the Urban dataset by different methods.

still rather large deviations from the original curves in Fig. 6. Bayesian-based methods, including HB, MIX_HB, and the proposed WT_HB methods, perform much better than others both in endmembers and their variations. Among them, the HB method shows a less accurate approximation to the endmember variability for the EM.2 and EM.4 in Fig. 7. Meanwhile, MIX_HB has a better performance on the Urban dataset in Fig. 6. However, it produces a relatively large error for the EM.3 in Fig. 7. The endmember curves of the proposed WT_HB method are obviously closer to the actual one both in shape and trend. Furthermore, WT_HB offers a more accurate approximation to the endmember variability than that of the other two methods, demonstrating the merits of the wavelet transform. However, it can be seen that, for the EM.4 in Fig. 6, WT_HB performs worse than other Bayesian-based methods. It can be interpreted that the EM.4 has fewer peaks and valleys with a flat curve, resulting in the failure of the wavelet transform. On the other hand, for the Jasper dataset in Fig. 7, the endmember variability curve obtained by WT_HB is much alike to the ground truth. In general, the endmember obtained by the WT_HB method is obviously closer to the real one with well-preserved peaks, valleys, and trends.

5) Abundance Estimation: Figs. 8 and 9 give the abundance map of the second and the first endmember for two datasets, respectively. In general, VCA, SISAL, PGMSU, ELMM, PLMM, OU, and HB methods have large abundance estimation error for the Urban dataset, especially at $t = 1$, 4, while, for the Jasper dataset, except VCA, SISAL, and PGMSU, other UN methods achieve better results at $t = 3$,

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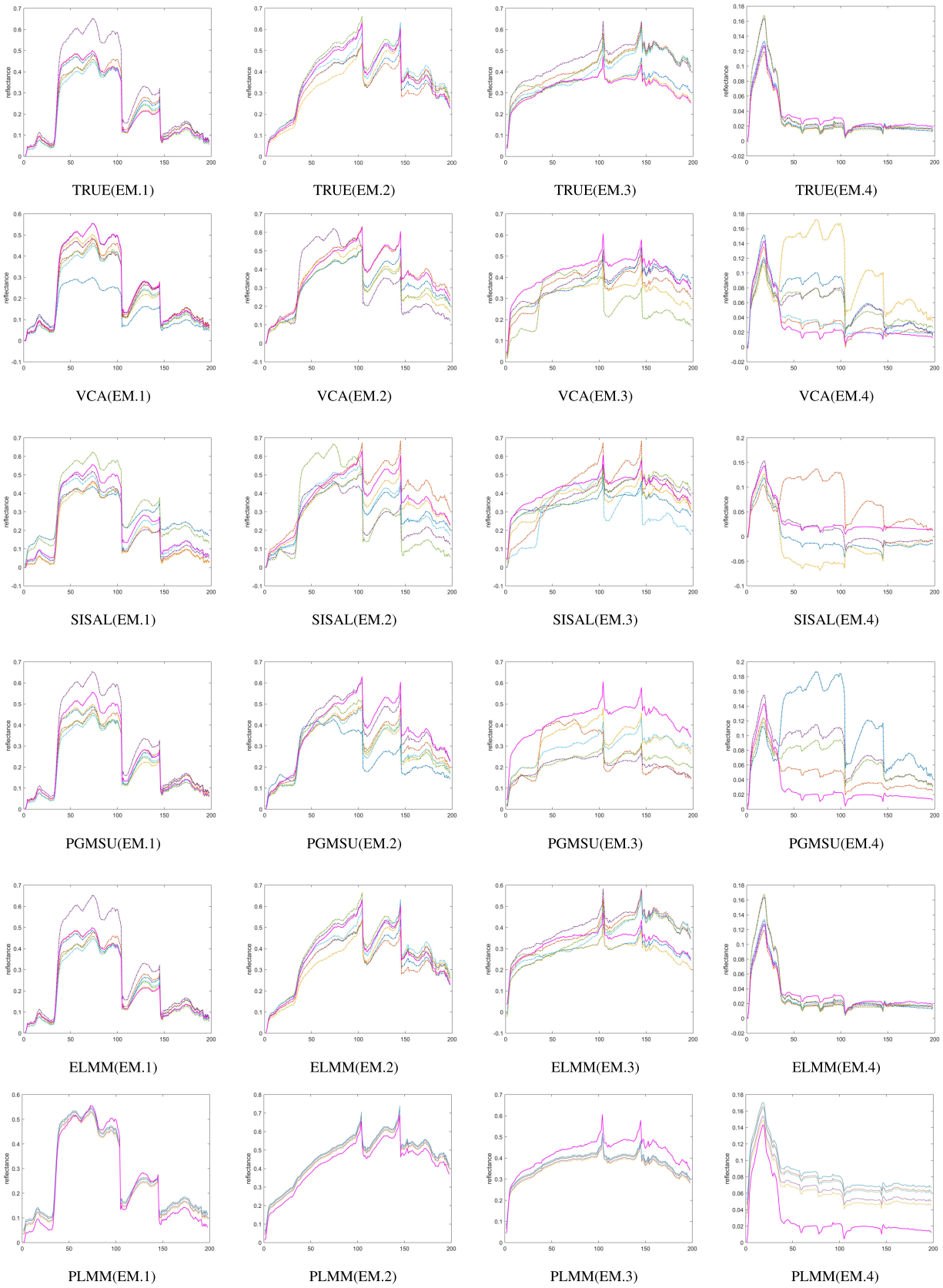


Fig. 7. Endmembers (m_r, purple lines) and their endmember variability ($m_r + dm_{r,t}$, dotted lines) recovered from the Jasper dataset by different methods.

5, 6 in Fig. 7. Specifically, VCA and PGMSU are better than SISAL for the Urban dataset in Fig. 8, in which VCA performs worse at $t = 3$ and PGUMSU performs worse at $t = 4$. For the Jasper dataset in Fig. 9, PGMSU has underestimate the

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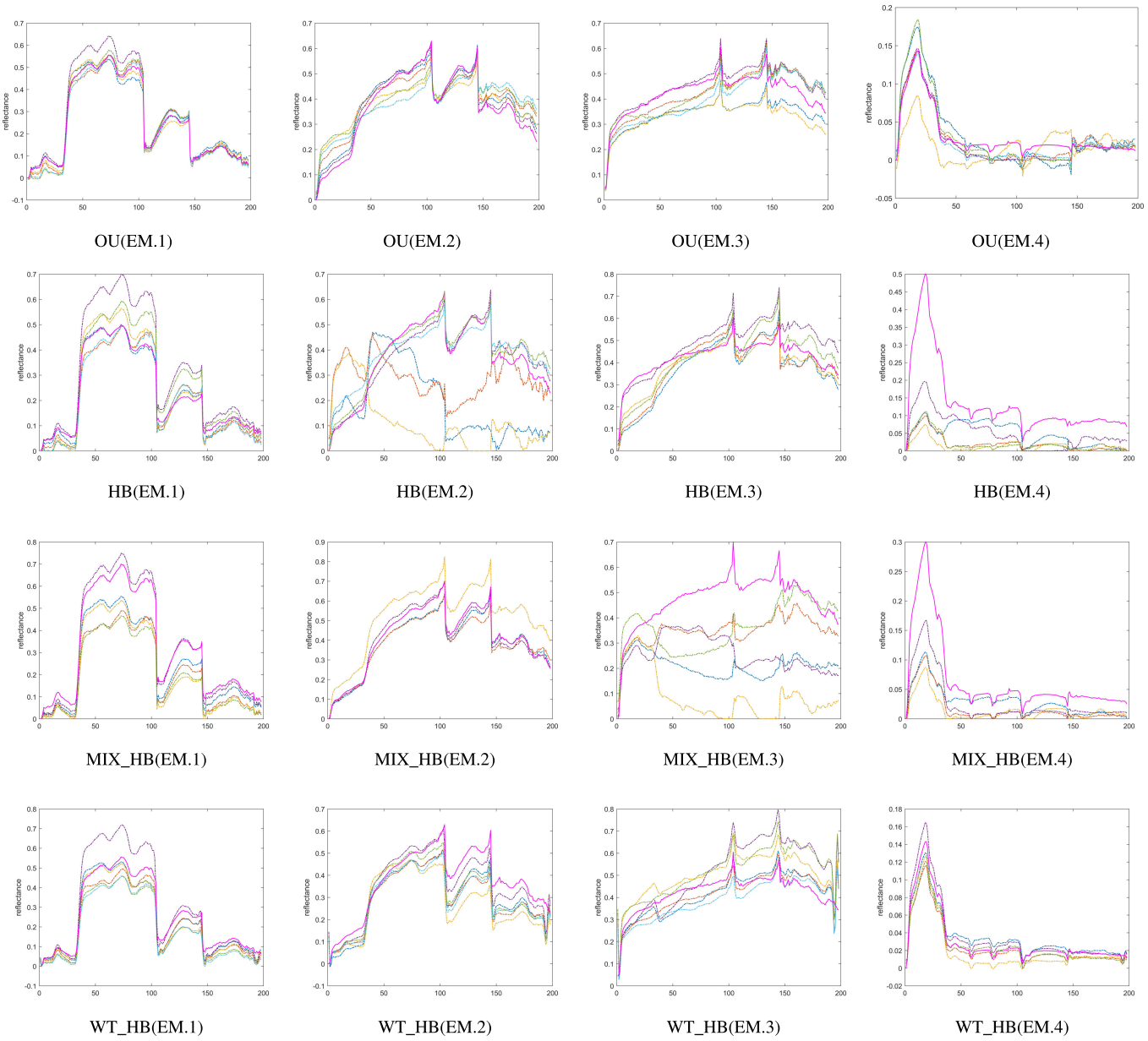


Fig. 7. *(Continued.)* Endmembers (m_r, purple lines) and their endmember variability ($m_r + dm_{r,t}$, dotted lines) recovered from the Jasper dataset by different methods.

abundance of the EM.1 with large errors, so dose SISAL, i.e., at $t = 1$. OU has a slight improvement compared with PLMM at $t = 4, 5, 6$. ELMM obtains more smooth abundances with values larger than that of the real one. The Bayesian-based methods achieve obviously satisfying approximation except for HB, in which the abundances are much greater than the real ones. For MIX_HB and WT_HB, they have similar results consistent with the endmember curves at Fig. 8, and their results are close to the ground truth at all time instants. For the Jasper dataset, HB misses the abundance information, and MIX_HB also has a slight distance from the real one at $t = 4$. Compared with HB and MIX_HB, the results obtained by the WT_HB method are much closer to the real results, especially at $t = 1, 2, 4$ in Fig. 9. Compared with HB and MIX_HB, the results obtained by the WT_HB method is much closer to the real results, especially at $t = 1, 2, 4$ in Fig. 9. Moreover, at $t = 4$ in Fig. 9, the real abundance has a large difference from other instants, resulting in a large estimation error. However, the WT_HB method has better recovery effects for the abundance at this moment. Meanwhile, from the results of numerical indexes, it is found that the WT_HB method can always generate the best approximation in abundance estimation.

6) Total Performance: The numerical indicators are shown in Tables IV and V. There are slight differences between the two datasets. VCA and SISAL have the worst indexes due to omitting the endmember variability. Although PGMSU has a relatively poor performance on SAM, its GMSE index gets to the lowest values for abundance estimation. ELMM and PLMM yield rather large SAM values, which may be caused by inappropriate parameters in the models. OU has

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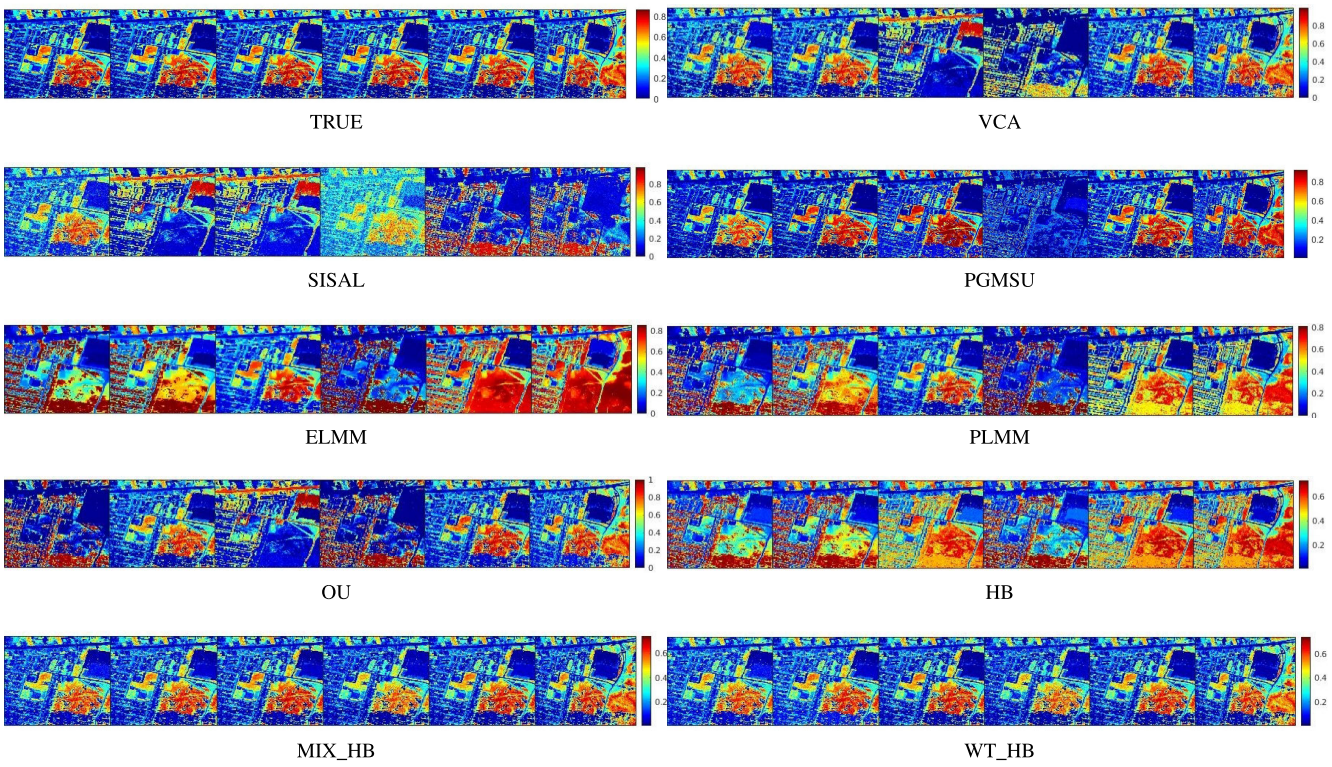


Fig. 8. Change of the second endmember's abundance graph at different times extracted from the Urban dataset.

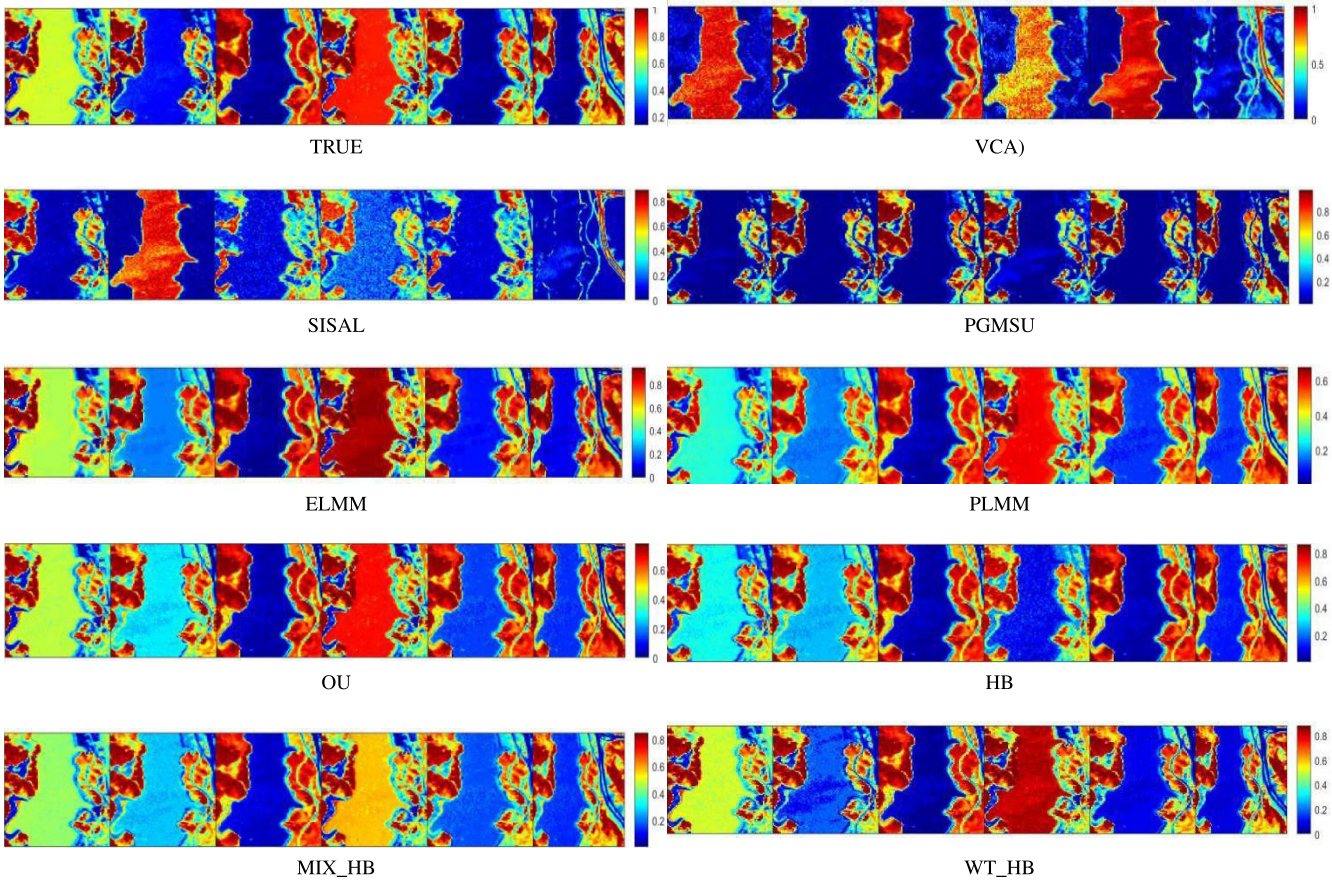


Fig. 9. Change of the first endmember's abundance graph at different times extracted from the Jasper dataset.

an improved SAM index compared with PLMM, and the GMAR(dM) for the Jasper dataset is best. PLMM produces the smaller REs due to the minimization of the loss function; MIX_HB has the best and second-best values on SAM

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TABLE IV
QUANTITATIVE RESULTS OF SIMULATED URBAN DATASET

Endmember	Methods	aSAM(M)	GMSE(A)	GMSE(dM)	RE
	VCA/FCLS	8.98	0.0428	/	6.04e-5
	SISAL/FCLS	9.97	0.0382	/	5.87e-5
	PGMSU	13.58	0.0048	/	0.0115
	ELMM	7.94	0.0433	6.03e-4	5.17e-5
R=4	PLMM	6.82	0.0334	**4.50e-4**	**8.02e-6**
	OU	7.56	0.0264	7.1e-4	4.92e-5
	HB	4.02	0.0315	0.0013	4.92e-5
	MIXHB	2.82	**0.0246**	0.0011	4.89e-5
	WTHB	**1.10**	0.0311	0.0029	4.80e-5
	VCA/FCLS	10.32	0.0342	/	8.36e-5
	SISAL/FCLS	9.83	0.0331	/	7.07e-5
	PGMSU	6.99	**0.0051**	/	0.0118
	ELMM	6.52	0.0414	0.0014	5.24e-5
R=5	PLMM	7.13	0.0380	**0.0011**	**8.58e-6**
	OU	4.34	0.0312	0.0017	6.37e-5
	HB	3.83	0.0257	0.0025	4.94e-5
	MIXHB	**3.40**	0.0204	0.0016	5.24e-5
	WTHB	6.24	0.0227	0.0057	4.33e-5
	VCA/FCLS	11.21	0.0418	/	7.21e-5
	SISAL/FCLS	10.32	0.0398	/	5.91e-5
	PGMSU	8.02	**0.0038**	/	0.0138
	ELMM	8.07	0.0388	0.0011	6.24e-5
R=6	PLMM	8.55	0.0384	**9.08e-4**	**6.01e-6**
	OU	6.43	0.0330	0.0011	4.98e-5
	HB	7.02	0.0221	0.0036	4.88e-5
	MIXHB	6.08	0.0198	0.0017	4.79e-5
	WTHB	**2.42**	0.0184	0.0124	4.37e-5

TABLE V
QUANTITATIVE RESULTS OF SIMULATED JASPER DATASET

Methods	aSAM(M)	GMSE(A)	GMSE(dM)	RE
VCA/FCLS	29.96	0.20	/	8.24e-6
SISAL/FCLS	31.38	0.1519	/	**8.16e-6**
PGMSU	16.41	0.2181	/	0.0373
ELMM	18.02	0.2510	0.002	0.0685
PLMM	16.10	0.20	0.002	9.87e-6
OU	12.60	0.25	**6.56e-4**	4.92e-5
HB	27.93	0.1749	0.02	4.85e-5
MIXHB	24.50	0.025	0.01	4.89e-5
WTHB	**11.97**	**0.0629**	0.0035	4.78e-5

and GMSE(A) for the Urban dataset. However, it performs worse on the Jasper dataset, which may be caused by its low performance on EM.3 on this dataset. The proposed WT_HB shows outstanding performance on quantity indexes. Its GMAE(A) and SAM are either best or second best, which shows the advantages of the wavelet-domain inherent features priors. Moreover, WT_HB shows good performance on the Jasper dataset with the lowest SAM, GMSE(A). Generally speaking, WT_HB achieves satisfactory UN results with the best estimation of the endmember. Meanwhile, in terms of abundance, there are also fewer errors than that of other compared methods. The same conclusion also can be drawn from the following real data experiments.

B. Real Data

The Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the Lake Tahoe region in California, USA, is used for the real data experiment. The SHS is composed of six frames of HSI obtained from 2014 to 2015 with 169 spectral bands and three endmembers (*Water, Soil*, and *Vegetation*), as shown in Fig. 10. The parameters used for the proposed approach are given in Table II.



Fig. 10. Sequence of real HS images taken in the same area at different times.

TABLE VI
TIMES (SECONDS)

Methods	Urban(R=4)	Japser
VCA/FCLS	**16.8**	**0.5646**
SISAL/FCLS	54	1.4044
PGMSU	896	69
ELMM	7.63e+3	1.11e+3
PLMM	2.01e+4	1.06e+3
OU 2	771	1.24e+2
HB 2	4.12e+4	5.10e+3
MIXHB	4.53e+4	4.30e+3
WTHB	7.68e+4	7.91e+3

Although there is no ground truth for the real data, some conclusions can also be drawn from Fig. 11. The spectral curves have the same phoneme as that of stimulated data. VCA and SISAL produce large deviations of each curve, while ELMM and PLMM have more compressed curves. PGMSU performs worse for the second endmember. OU, HB, and WT_HB can present the variabilities well. Specifically, the WT_HB has a good tradeoff between the trend and the deviation, i.e., the first and third endmembers are relatively stable with changes, and the second endmember reflects the trend (solid line) of all the variabilities (dotted lines) well.

Figs. 12–14 show the abundance map obtained by these methods. VCA produces a large error in the first endmember at $t = 2$, while WT_HB has a good estimation at the same time. PGMSU underestimates the first endmember. For the second endmember (*Water*) at $t = 3, 4$, it has a little ratio; VCA and SISAL underestimate the ratio of *Water*, while PGMSU overestimates the second and the third endmembers. ELMM cannot detect the first and third endmembers. At $t = 1, 2, 3, 4$, HB underestimates the endmember Soil, while, at $t = 3, 5, 6$, OU overestimates it. At $t = 1, 2, 3, 4$, WT_HB has different estimates with HB and MIX_HB, which is much closer to the real one. OU PLMM and WT_HB have more similar figures on the second endmember. However, there are large differences among HB, MIX_HB, and WT_HB at $t = 3, 4$. For the third endmember *Soil*, VCA cannot detect it well, and all the methods yield inconsistent estimates with time varies. Compared with the real endmember *Soil*, the WT_HB represents the abundance well, and the obtained abundance map is roughly equivalent to the distribution of substances in the real image.

C. Time Complexity Analysis

Finally, the time consumption is compared on Urban and Jasper datasets. It can be seen from Table VI that VCA is the fastest algorithm, which is why it is often used as an initial value in many UN methods. PGMSU is a learning-based method; once the parameter is learned, the test gets very fast. PLMM, ELMM, and OU have a consideration time computation compared with VCA. However, they are faster

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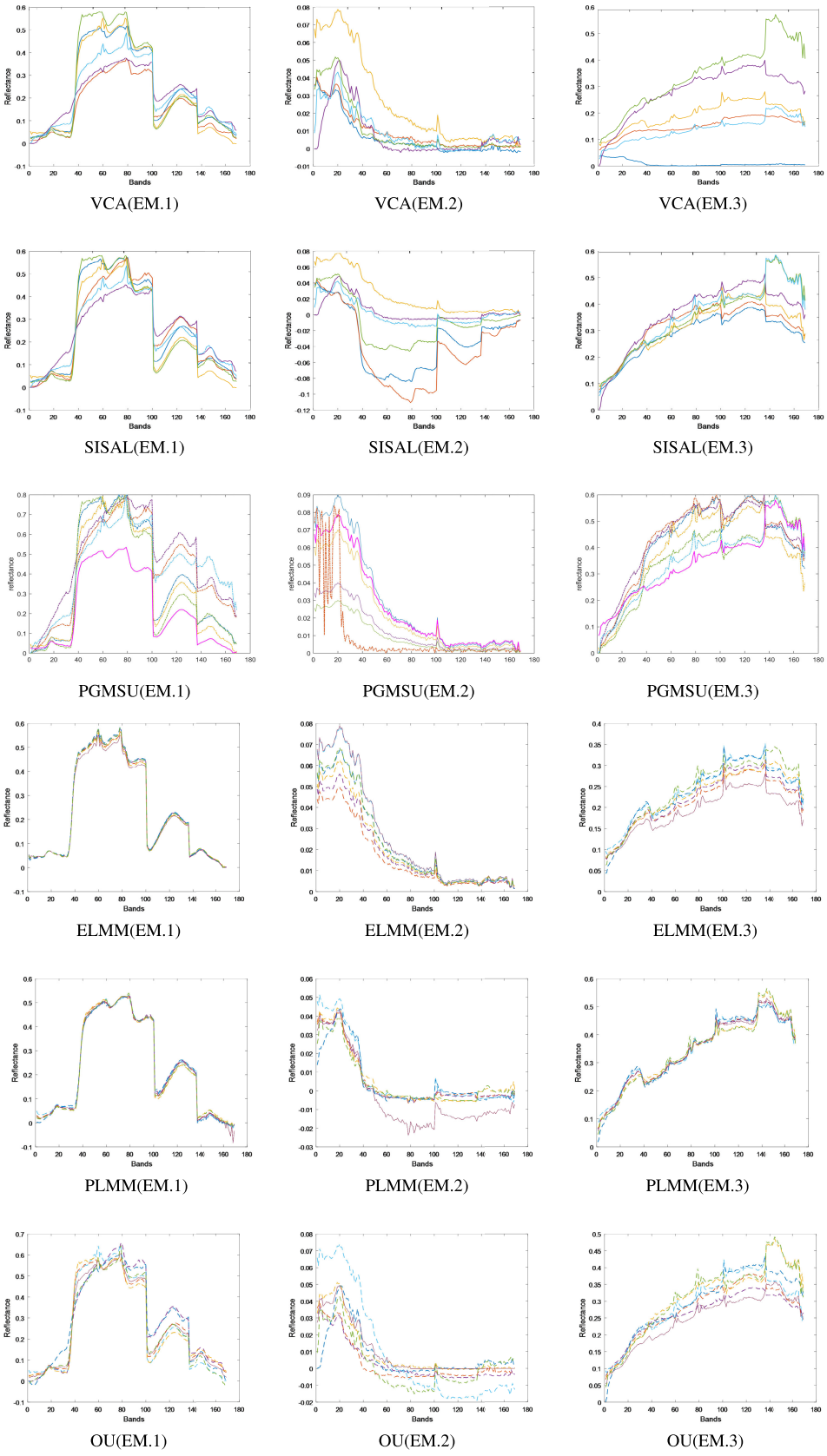


Fig. 11. Endmembers (m_r, purple lines) and their endmember variability ($m_r + dm_{r,t}$, dotted lines) recovered from the real dataset by different methods.

than the Bayesian-based methods. HB, MIX_HB, and WT_HB employ the MCMC algorithm, which needs a long time to burn-in for a Markov Chain to get to a stable state, so the time complexity is very high. Moreover, WT_HB is computed

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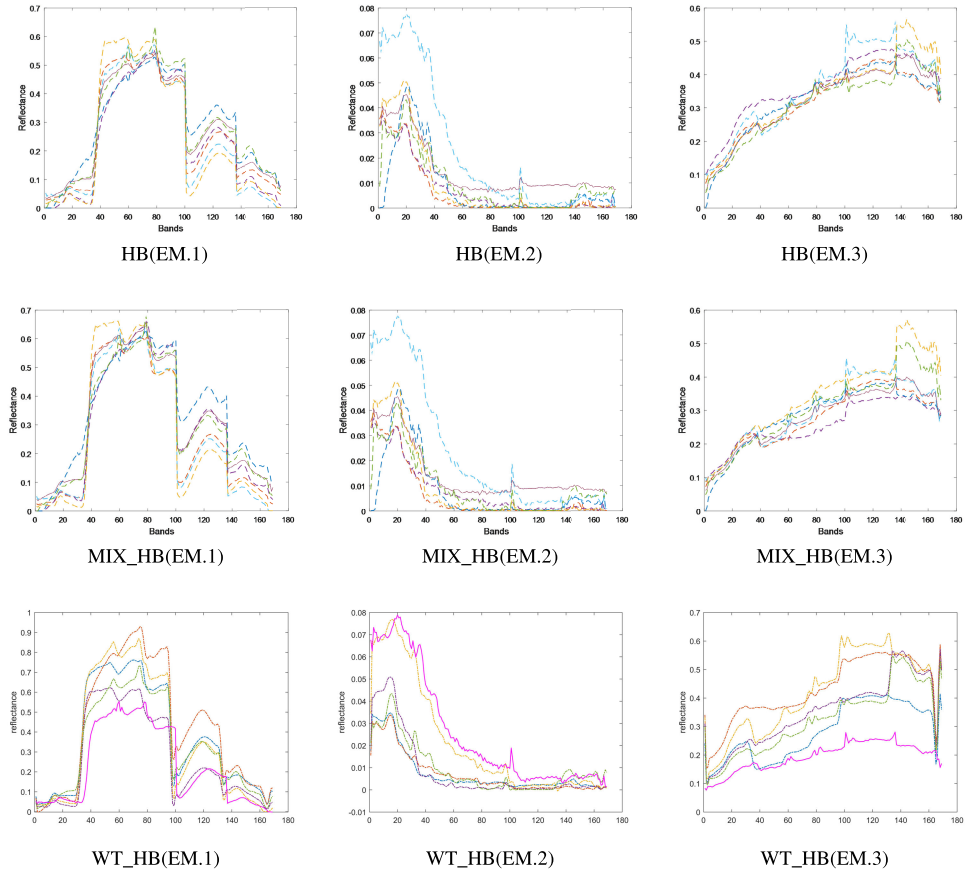


Fig. 11. *(Continued.)* Endmembers (m_r, purple lines) and their endmember variability ($m_r + dm_{r,t}$, dotted lines) recovered from the real dataset by different methods.

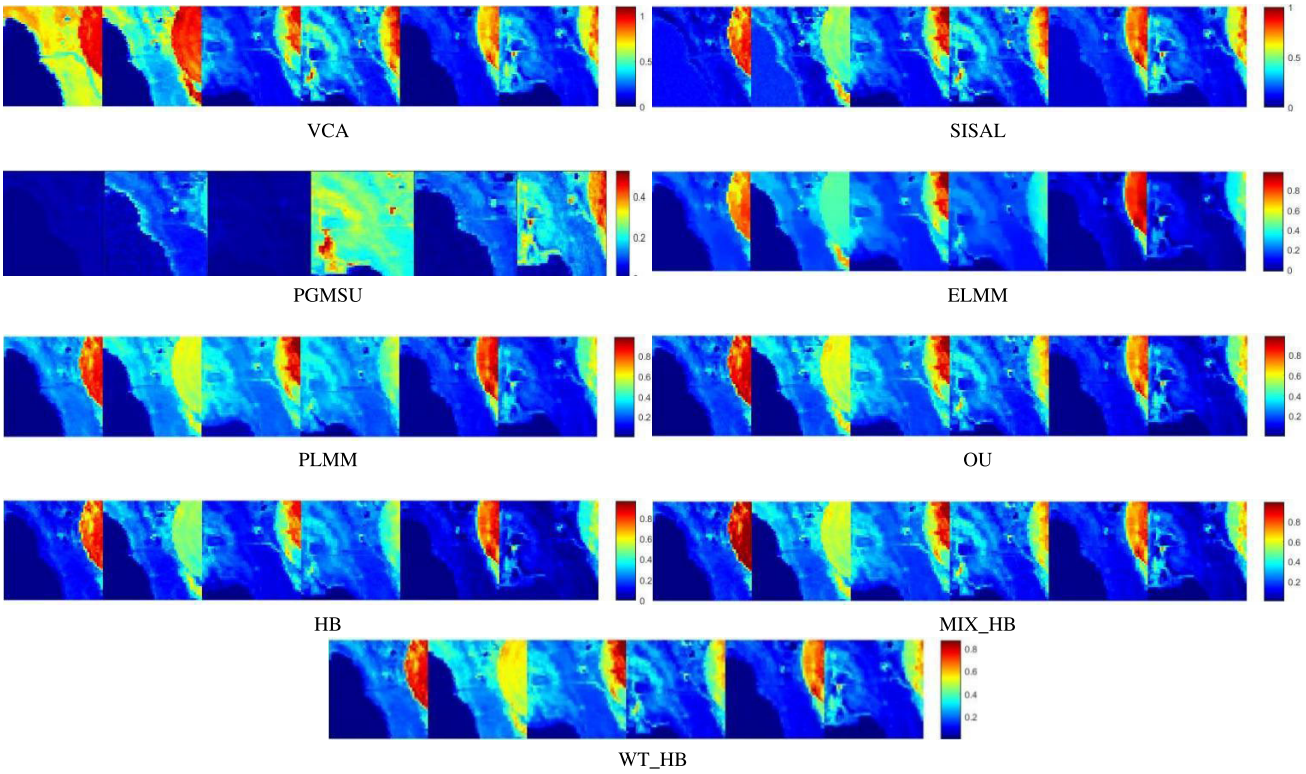


Fig. 12. Change of the first endmember's (*Veg.*) abundance map at different times extracted from the real dataset.

based on wavelet transform, so it takes more time than HB and MIX_HB methods. An efficient strategy for the wavelet transform and sampling algorithm is encouraged to accelerate the UN method.

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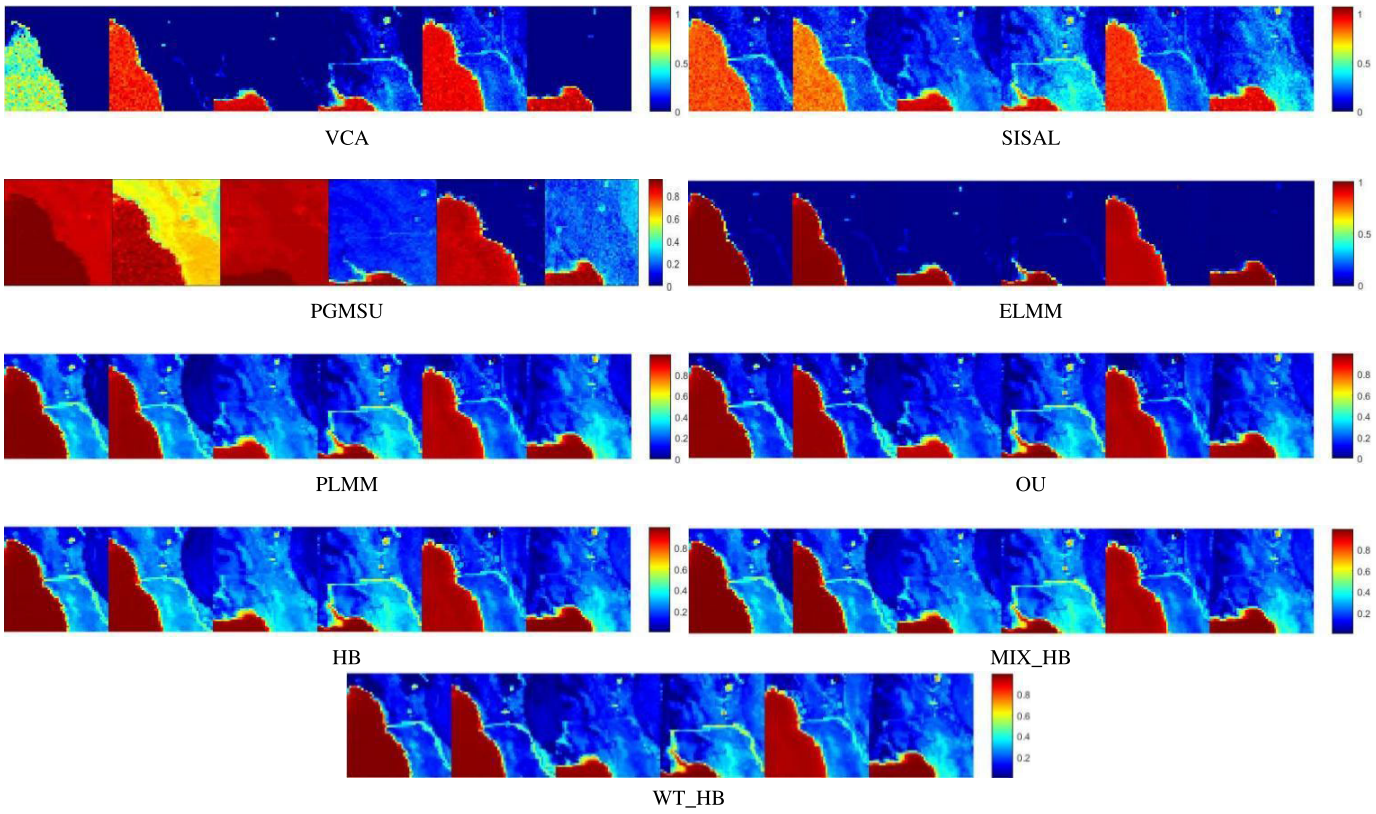


Fig. 13. Change of the second endmember's (*Water*) abundance map at different times extracted from the real dataset.

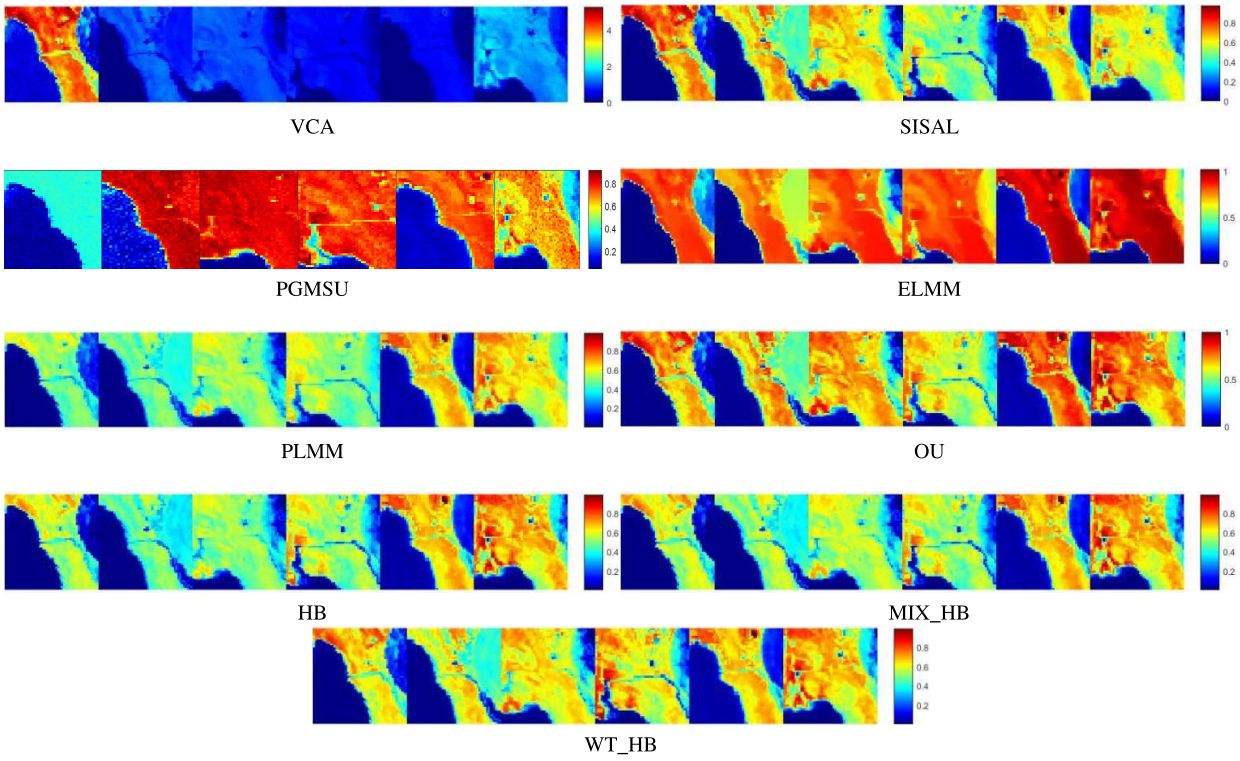


Fig. 14. Change of the third endmember's (*Soil*) abundance map at different times extracted from the real dataset.

VI. Conclusion

An SHS contains spectral variability due to various imaging environments. To make accurate estimations of the endmember and its variability, a new SHS UN in the wavelet domain is proposed. The advantage of wavelet transform in extracting the intrinsic feature of the spectral curves is fully exploited.

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Furthermore, combined with the Bayesian method, the high- and low-frequency wavelet coefficients of endmembers and their variation are given different priors based on the feature of the curves in the wavelet domain. Compared with the state-of-the-art UN methods, the proposed UN method has certain advantages in the estimation of the endmembers and their abundances. Future work will include the intrascale and interscale relationships of the wavelet coefficient under different degradations. Moreover, physical models, such as Hapke, or the digital surface model can be taken into consideration.

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